UFCFK1-15-0 - Fundamentals of data science 25feb_1

# Take home assessment 40% (PSA)

## Documentation 1

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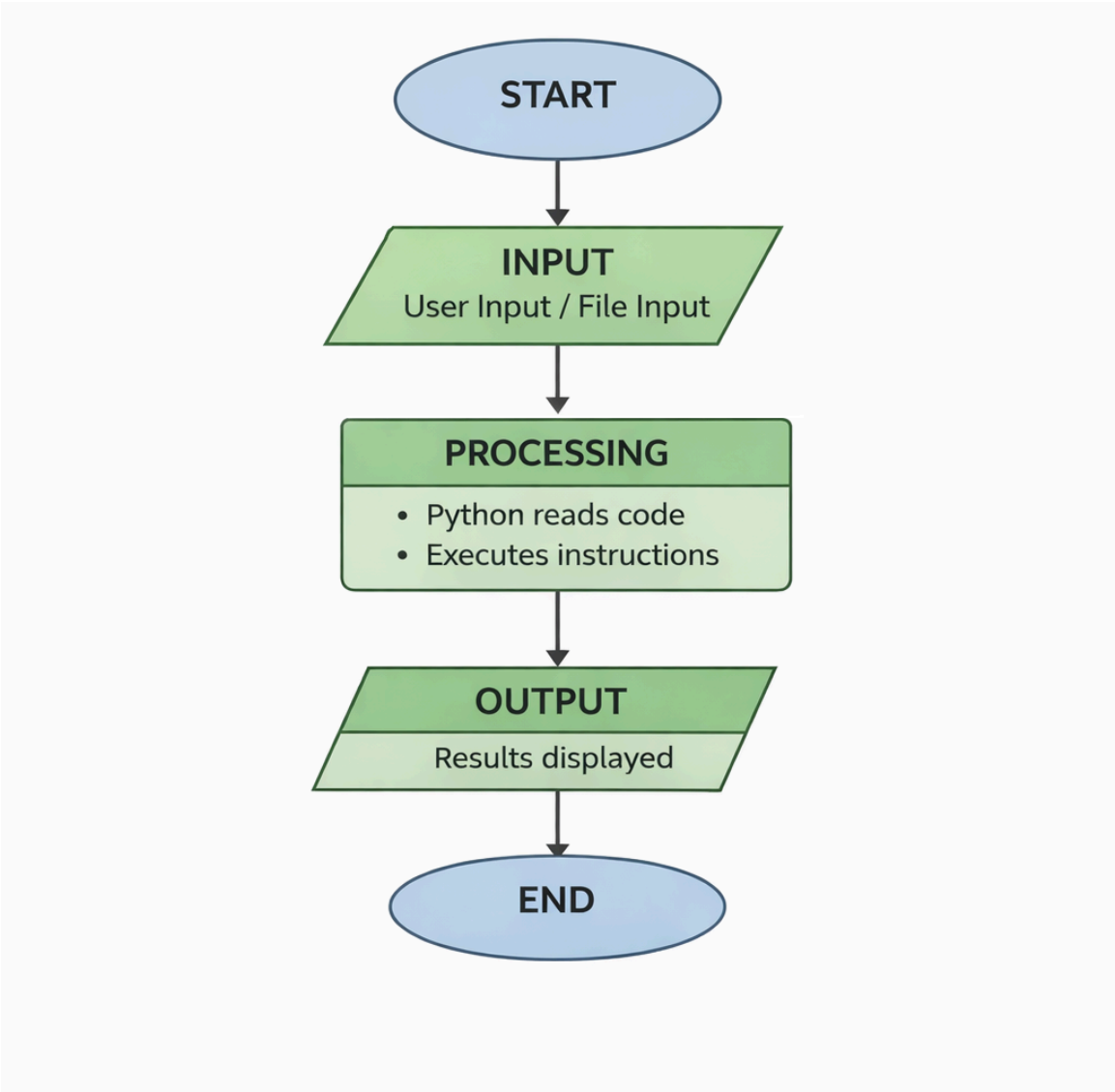
Section : A

GIT : https://github.com/999-ztj/FODS_assignments

## Question 1

The working of python



## Question 2

Program Description:

*This program performs basic arithmetic operations on two numbers.*

Screenshot:

```
Enter first number: 6
Enter second number: 3
Addition: 9
Multiplication: 18
Division: 2.0
Modulus: 0
Exponentiation: 216
```

## Question 3

Program Description:

*This program checks whether a number is positive/negative and even/odd.*

Screenshot:

```
Enter a number: 2
Positive and Even
```

```
Enter a number: 3
Positive and Odd
```

```
Enter a number: -6
Negative and  Even
```

```
Enter a number: -17
Negative and Odd
```

```
Enter a number: 0
Zero
```

## Question 4

Program Description:

```
This program performs cube, cube root, and logarithmic operations.
```

Screenshot:

```
Enter a number: 8
Cube: 512.0
Cube root: 2.0
Natural log: 2.0794415416798357
Log base 2: 3.0
Power 6: 262144.0
```

## Question 5

Program Description:

```
This program calculates total, average, percentage and grade.
```

Screenshot:

```
Enter marks for 6 subjects (out of 100):
  Subject 1: 50
  Subject 2: 90
  Subject 3: 70
  Subject 4: 75
  Subject 5: 67
  Subject 6: 17

===== Result =====
Marks          : [50.0, 90.0, 70.0, 75.0, 67.0, 17.0]
Total          : 369.0 / 600
Average        : 61.50
Percentage     : 61.50%
Grade          : Second Division
```

## Question 6

Program Description:

```
This program finds prime numbers in a given range.
```

Screenshot:

```
Enter start: 1
Enter end: 100
2
3
5
7
11
13
17
19
23
29
31
37
41
43
47
53
59
61
67
71
73
79
83
89
97
Count: 25
Sum: 1060
```

## Question 7

Program Description:

```
This program finds numbers divisible by 9 but not by 6.
```

Screenshot:

```
Enter start: 1
Enter end: 100
9
27
45
63
81
99
```

1000 - 2500

## Question 8

Program Description:

```
This program calculates the factorial of a positive integer.
```

Screenshot:

```
Enter a number: 5
Factorial: 120
```

```
Enter a number: -2
Invalid input! Please enter a positive integer.
```

## Question 9

Program Description:

```
This program sums positive and negative numbers separately.
```

Screenshot:

```
Enter number: 17
Continue? (y/n): y
Enter number: 19
Continue? (y/n): y
Enter number: -17
Continue? (y/n): y
Enter number: -19
Continue? (y/n): n
Positive sum: 36
Negative sum: -36
```

## Question 10

Program Description:

```
This program is a number guessing game.
```

Screenshot:

```
Enter your guess: 50
Too high
Enter your guess: 25
Too high
Enter your guess: 15
Too high
Enter your guess: 10
Too high
Enter your guess: 5
Too high
Enter your guess: 3
Too high
Enter your guess: 2
Correct!
```